

ST ANDREW'S JUNIOR COLLEGE



JC2 Preliminary Examination

8873/02

1 Sep 2021

0800 – 1000

2 Hours

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A [60 marks]

Answer **all** the questions.

Section B [20 marks]

Answer **one** question.

The use of an approved calculator scientific calculator is expected, where appropriate.

At the end of the examination, fasten all your work securely together.

You are reminded of the need for good English and clear presentation in your answers. The number of marks is given in brackets [] at the end of each question or part question.

For Examiners' use only:

Section A			
Question 1	18	Question 2	9
Question 3	5	Question 4	10
Question 5	10	Question 6	8
Section B			
Question 7/8	20		
TOTAL			80

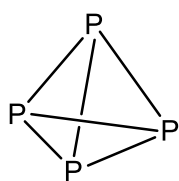
This document consists of 24 pages.

Section A (60 marks)

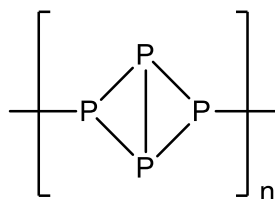
Answer **all** the questions in this section in the space provided.

- 1 Phosphorus and sulfur are elements in Period 3 of the Periodic Table and each can exist in various allotropic forms.

Phosphorus can exist as white phosphorus and red phosphorus. White phosphorus exists as a tetrahedron with a bond angle 60° while red phosphorus exists as a polymeric chain of regular tetrahedrons.



white phosphorus



red phosphorus

- (a) The bond angle in red phosphorus is 80° .

By considering the number of bond pairs and lone pairs, suggest why white phosphorus is less stable than the red phosphorus.

[2]

- (b) The boiling points of the two chlorides of phosphorous are shown in the table below.

compound	boiling point / $^\circ\text{C}$
PCl_3	76.1
PCl_5	166.8

- (i) With reference to the structure and bonding, explain the difference in boiling points between the two chlorides of phosphorous.

[2]

- 1 (b) (ii) PCl_3 is able to react with BF_3 . State the type of bond formed and draw a diagram to illustrate the shape of the product obtained. [2]

Type of bond

- (c) A sample of pure sulfur which contains only ^{32}S and ^{33}S has a relative atomic mass of 32.08.

- (i) Define the term *relative atomic mass* of an element. [1]

- (ii) Calculate the percentage abundances of both ^{32}S and ^{33}S in this sample of sulfur. [1]

- (d) When sulfur is heated under pressure with chlorine, the major product is SCl_2 . Under suitable conditions, SCl_2 hydrolyse in water to produce sulfur, SO_2 and HCl . During the reaction, 35 kJ mol^{-1} of energy is released.

- (i) Constructing an equation for the hydrolysis of SCl_2 . [1]

- 1 (d) (ii) Hence, explain in terms of oxidation numbers, what happens to the sulfur in this reaction and state the type of reaction. [2]

- (iii) Draw an energy profile diagram for the reaction in (e). Label the activation energy E_a , and the enthalpy of reaction ΔH . [1]



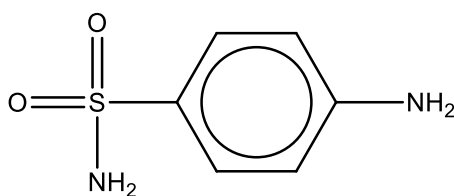
- (e) Sulfur can react with magnesium to form magnesium sulfide, MgS . [1]
- (i) Draw a 'dot-and-cross' diagram for magnesium sulfide.

- (ii) Magnesium sulfide dissolves in water. [2]
- Use your knowledge of the Periodic Table to predict a value for the pH of the solution formed. Explain how you arrived at your answer.

- 1 (e) (iii) Explain why a sulfur atom has a smaller atomic radius than a magnesium atom but a sulfide ion has a larger ionic radius than a magnesium ion. [3]

[Total: 18 marks]

- 2 Sulfanilamide, is an anti-bacterial drug, which had been used safely in tablet and powder form. However, in 1937, Elixir sulfanilamide, a medicine which was synthesised using diethylene glycol as a solvent, poisoned and killed more than 100 people because of acute kidney failure. It was found out later that it was the solvent that was poisonous.



sulfanilamide



The maximum daily dosage of sulfanilamide is 0.08 g per kg of patient. An average adult may be assumed to weigh about 70 kg.

- (a) Is it safe for an infected person to consume 15 capsules in a single day? Justify with calculations. [1]

- (b) (i) Assuming sulfanilamide is a monoacidic base, write an equation to show how sulfanilamide react with aqueous HCl. [1]
You may use RNH_2 to represent sulfanilamide.

- (ii) Hence, explain why sulfanilamide is soluble in acids but not in water. [1]

- (c) 0.15 g of an antibiotic power containing sulfanilamide was dissolved in 100 cm³ of distilled water to form an aqueous solution.

A 25.0 cm³ aliquot was transferred into a conical flask, in which 10.0 cm³ of 0.0250 mol dm⁻³ KBrO₃ was added. About 10 g of solid KBr was then added.

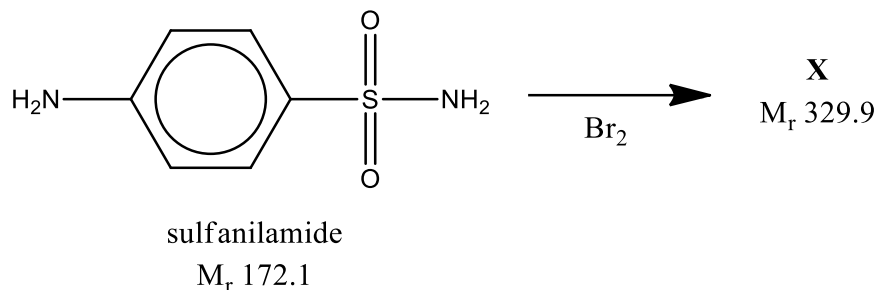
BrO₃⁻ reacts with bromide according to the equation:



- (i) Calculate the number of moles of Br₂ formed. [1]

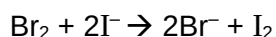
- 2 (c) The bromine formed then reacts with the sulfanilamide to form **X**. In this reaction, the bromine atom substitutes the H in the benzene ring.

Note that the following equation is not balanced.

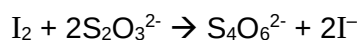


- (ii) By comparing the M_r of sulfanilamide and **X**, deduce whether **X** is a mono-brominated or di-brominated compound. Show your working. [1]

- (iii) After 15 minutes, an excess of KI was added.



The liberated iodine was then titrated with 15.00 cm³ of 0.08500 mol dm⁻³ Na₂S₂O₃.



Calculate the number of moles of bromine, which reacted with the KI. [1]

- (iv) Using your answers to (c)(i) and (c)(iii), calculate the number of moles of sulfanilamide which reacted with the bromine. [1]

- 2 (c) (v) Assuming the mole ratio of bromine:sulfanilamide is 1:1, calculate the percentage by mass of sulfanilamide in the sample. [2]

[Total: 9 marks]

- 3 (a) The halogens, found in Group 17 of the Periodic Table, and their compounds are useful laboratory reagents.

The table below shows some properties of the halogens and their compounds.

Halogen, X_2	chlorine	bromine	iodine
standard enthalpy change of vapourisation $(\Delta H_v^\ominus)$ of X_2 / kJ mol^{-1}	0	+15	+30
electronegativity	3.2	3.0	2.7
standard enthalpy change of reaction $(\Delta H_r^\ominus)$ for thermal decomposition of HX / kJ mol^{-1}	+183	+103	+11

By quoting relevant data from the *Data Booklet*, explain the standard enthalpy change of reaction $(\Delta H_r^\ominus)$ for thermal decomposition of the hydrides of the Group 17 elements. [2]

- 3 (b)** Astatine, At, is found below iodine in Group 17 of the Periodic Table. Using your knowledge of the chemistry of Group 17, predict the changes (if any) that you may observe when the following reagents are mixed.

If nothing is formed, write “no reaction” and where there is a reaction predicted, write an ionic equation with state symbols.

- (i)** aqueous bromine and aqueous sodium astatide **[1]**

- (ii)** astatine and aqueous sodium chloride **[1]**

- (c)** When sodium iodide reacts with concentrated sulfuric acid, iodine is formed as one of the products.

When sodium chloride reacts with concentrated sulfuric acid, chlorine is **not** formed as one of the products.

Suggest why when concentrated sulfuric acid is reacted with these halide salts, iodine is formed but chlorine is not. **[1]**

[Total: 5 marks]

- 4 (a)** The Haber-Weiss reaction generates highly reactive free radicals from H_2O_2 . A free radical is formed when the O–O covalent bond in H_2O_2 breaks. Explain why the free radical is highly reactive. **[1]**

- 4 (b) Hydrogen peroxide can undergo disproportionation reaction in the presence of the Fe^{2+} catalyst.



The uncatalysed reaction has an activation energy of $+49 \text{ kJ mol}^{-1}$.

- (i) Define the term *activation energy*. [1]

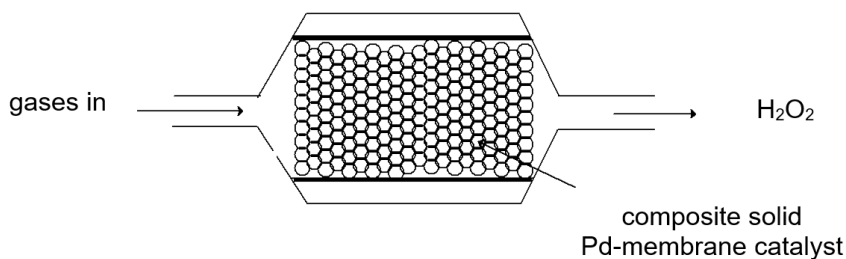
- (ii) Draw and label a Boltzmann distribution curve for the molecular energies of a sample of a H_2O_2 mixture.

On your diagram, draw a line to represent the activation energy. Label this line E_a . [2]

- (iii) On your Boltzmann distribution curve in **b(ii)**, show the change that would happen when a catalyst is added to the reaction mixture given that the temperature and pressure are kept constant. Label the change X. [1]

- 4 (c) H_2O_2 is in high demand in chemical industries as it is considered an environmentally-friendly oxidising agent.

In recent times, a novel composite solid palladium membrane catalyst for the direct oxidation of H_2 to H_2O_2 has been used.



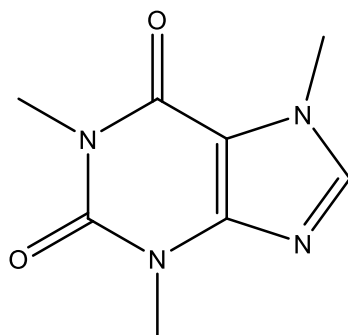
- (i) Suggest why it is environmentally-friendly to use H_2O_2 as an oxidising agent. [1]

- (ii) State what is meant by the term *heterogeneous* when it is applied to a Pd catalyst in the oxidation reaction of H_2 to H_2O_2 . [1]

- (iii) Outline, in three stages, the mode of action as Pd catalyses the oxidation reaction of H_2 . [3]

[Total: 10 marks]

- 5 Holly berries contain bitter compounds such as alkaloids, which make them mildly toxic if eaten. One alkaloid culprit is theobromine, which is also found in small amounts in chocolate.



theobromine

Figure 5.1

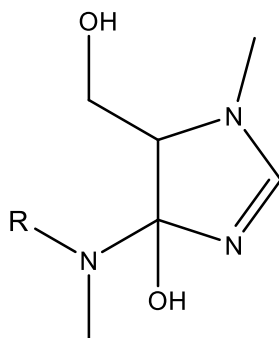
- (a) (i) State the molecular formula of theobromine. [1]

-
- (ii) Theobromine can be hydrolysed.

On **Figure 5.1**, draw lines through all the bonds that will be broken when theobromine is completely hydrolysed. [2]

- (iii) Draw the structure of the acidic hydrolysis product with the biggest M_r . [1]

Compound **X** is used in the synthesis of theobromine.



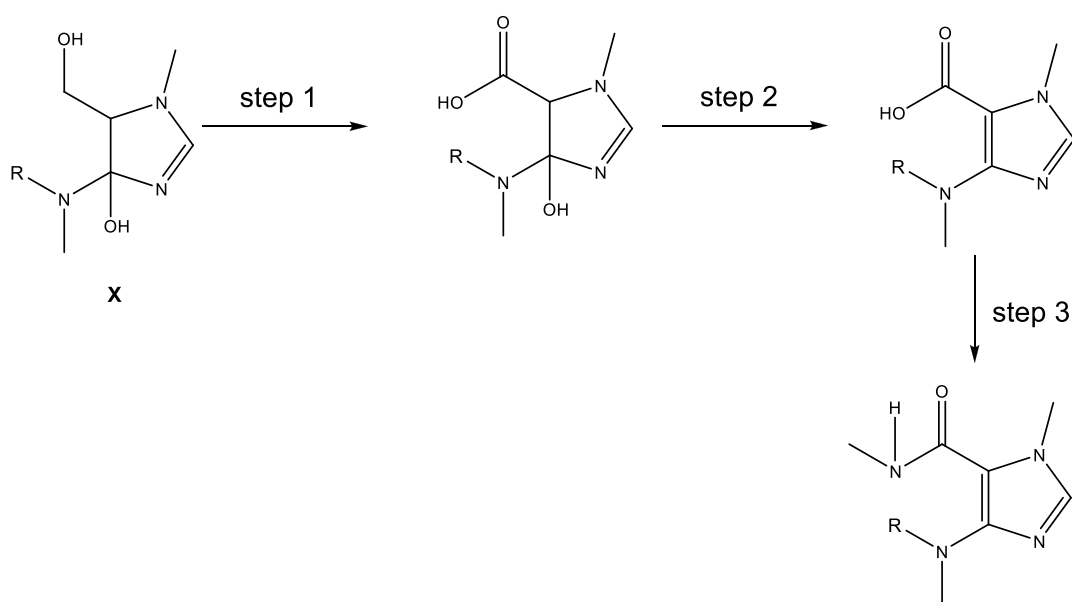
Compound **X**

- 5 (b) Use the table of characteristic infra-red absorption frequencies in the *Data Booklet* to answer this question.

Use the table to identify an infra-red absorption range that will be shown by compound **X** but not by theobromine [1]

..... cm^{-1}

- (c) The following method was proposed by a student to show part of the synthesis of theobromine from compound **X**.



State the reagents and conditions for steps 1 to 3.

[3]

Step 1:

.....

Step 2:

.....

Step 3:

.....

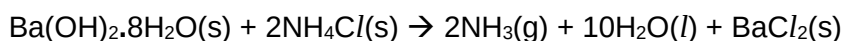
- 5 (d) There are four isomeric bromoalkanes with the formula C_4H_9Br . They can all [2]
form alkenes when undergoing elimination reactions.

Name and draw the **displayed formula** of the isomer, which forms an alkene that exhibits *cis-trans* isomerism.

Name of isomer.....

[Total: 10 marks]

6. Solid hydrated barium hydroxide, $Ba(OH)_2 \cdot 8H_2O$ reacts with solid ammonium chloride at room temperature according to the following equation.



	$\Delta H_f / \text{kJmol}^{-1}$
$Ba(OH)_2 \cdot 8H_2O(s)$	-3345
$NH_4Cl(s)$	-314
$NH_3(g)$	-46
$H_2O(l)$	-286
$BaCl_2(s)$	-859

Table 6.1

- (a) Use the equation and the data given in **Table 6.1** to calculate the enthalpy change of reaction for the reaction. [2]

- 6 (b) (i) Define the term *lattice energy*. [1]

- (ii) State how the magnitude of the lattice energy of $\text{BaCl}_2(\text{s})$ compares to that of $\text{NH}_4\text{Cl}(\text{s})$. Explain your answer. [2]

- (c) Barium hydroxide is a strong base.
Calculate the pH of $0.015\text{mol dm}^{-3} \text{Ba}(\text{OH})_2$. [2]

- (d) *Use of the Data Booklet is relevant to this question.*

$^{137}_{56}\text{Ba}$ is able to undergo a process known as alpha(α)-decay in which a helium nucleus, ^4_2He , and another species is emitted.

Complete the equation for the alpha(α)decay of $^{137}_{56}\text{Ba}$. [1]

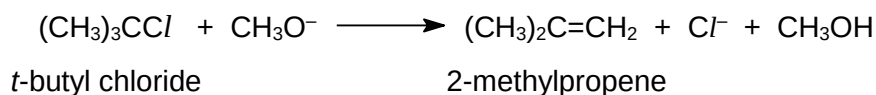


[Total: 8 marks]

Section B

Answer **one** question from this section in the spaces provided.

7. (a) *t*-butyl chloride reacts with alcoholic sodium methoxide, $\text{CH}_3\text{O}^-\text{Na}^+$, to form 2-methylpropene.



The kinetics of this reaction was studied and the following results were obtained.

experiment	$[(\text{CH}_3)_3\text{CCl}]$ /mol dm ⁻³	$[\text{CH}_3\text{O}^-]$ /mol dm ⁻³	Initial rate /mol dm ⁻³ min ⁻¹
1	0.0500	0.0400	4.00×10^{-6}
2	0.0300	0.0400	2.40×10^{-6}
3	0.0200	0.0300	1.60×10^{-6}

Table 7.1

- (i) State the type of reaction when *t*-butyl chloride reacts with alcoholic sodium methoxide. [1]

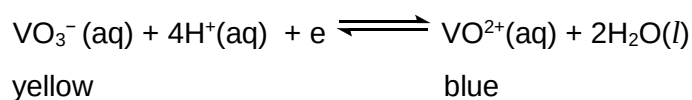
- (ii) Using the data in **Table 7.1**, deduce the order of reaction with respect to $[(\text{CH}_3)_3\text{CCl}]$ and $[\text{CH}_3\text{O}^-]$. [2]

- 7 (a) (iii) Using Experiment 1, calculate a value for the rate constant. Include units in your answer. [2]

units = _____

- (iv) Suggest a chemical test to distinguish *t*-butyl chloride and 2-methylpropene. State any observations you would make with each compound. [2]

- (b) Aqueous sodium vanadate(V) contains VO_3^- ions. These vanadate(V) ions exist in equilibrium with vanadate(IV) ions as shown by the expression below.



- (i) State the full electronic configuration of the vanadium ion in VO_3^- . [1]

- (ii) Write an expression for the equilibrium constant, K_c , for this reaction. [1]

- (iii) A solution is prepared in which the initial concentration of VO_3^- ions and H^+ ions are both 1.00 mol dm^{-3} . After equilibrium is reached, the concentration of VO^{2+} ions is $0.125 \text{ mol dm}^{-3}$.

Calculate the value of K_c and state its units. [3]

- 7 (b) (iv) If water is added to the vessel containing VO_3^- (aq) ions and VO^{2+} (aq) ions in equilibrium, the green solution becomes more yellow. Explain the observation. [2]

- (c) In 2021, Singapore built one of the world's biggest floating solar farms. 122 000 solar panels on the surface of Tengah Reservoir will power PUB's water treatment plants. The panels, made of glass and metal, are supported by floats made of food-grade certified high density polyethylene, HDPE, to ensure that water quality is not compromised. They are also UV-resistant and can withstand intense sunlight.

<https://www.straitstimes.com/multimedia/graphics/2021/05/singapore-largest-solar-farm-water/index.html?shell>

- (i) Describe the structure of HDPE. [2]

- (ii) A material is food grade certified if the material is safe to come into direct contact with food products. Suggest why HDPE is food grade certified. [1]

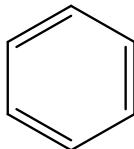
- 7 (c) (iii) Suggest why HDPE is used over LDPE when used as a float to support the heavy weight of the solar panels. [1]

- (iv) Draw labelled diagrams to illustrate the difference between HDPE and low density polyethylene, LDPE. [2]

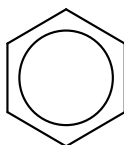
[Total: 20 marks]

8. In the past, several different structures have been suggested for benzene.

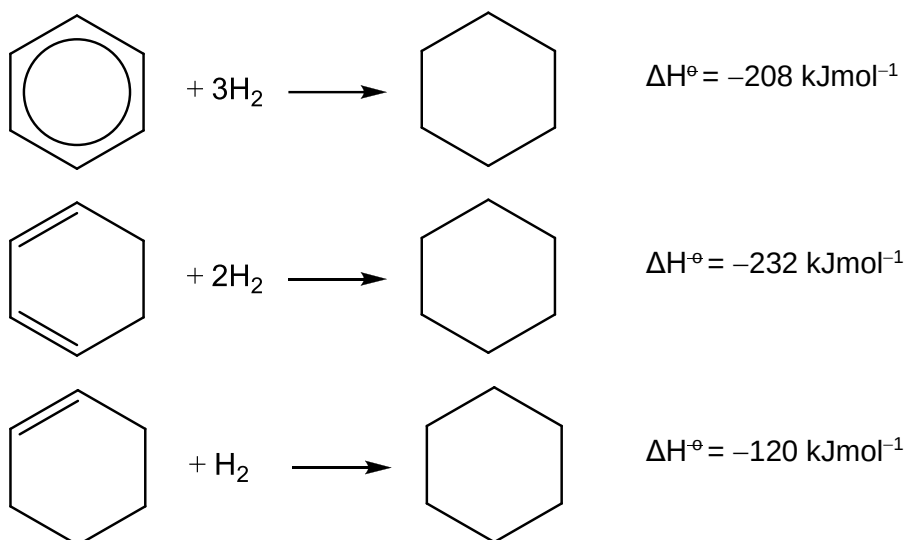
One of the suggested structures, which has been proved to be incorrect, was a compound containing three fixed double bonds, cyclohexa-1,3,5-triene.



The currently accepted structure for benzene can be represented as follows.

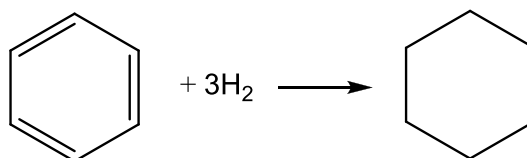


Benzene and the unsaturated cyclic hydrocarbons can be reduced to cyclohexane as shown in the following equations.



- (a) (i) Estimate the value of the standard enthalpy change of the reaction below when cyclohexa-1,3,5-triene is reduced to cyclohexane.

[1]



Estimated value =

- 8 (a) (ii) Explain your answer to (a)(i). [1]

- (iii) Explain why benzene does not exist as the cyclohexa-1,3,5-triene, using the data in the question and your answer to (a)(i) [2]

- (iv) Name the other reagent needed for the reduction of cyclohexa-1,3,5-triene [1]

- (b) (i) Describe the bonding in benzene in terms of orbital overlap. You may draw a diagram to illustrate your answer. [3]

- 8 (b) (ii) Write an equation to represent the *standard enthalpy change of combustion of liquid benzene*. [1]

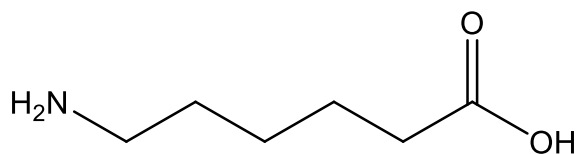
- (iii) Use appropriate bond energy values from the *Data Booklet* to calculate a value for the enthalpy change of combustion for benzene. [2]

Aqueous silver nitrate can be used to distinguish between bromobenzene and bromoethane.

- (iii) With the aid of suitable equations, describe how silver nitrate can be used to distinguish between bromobenzene and bromoethane. State any observations. [2]

- (iv) Explain why the substitution reaction with bromoethane occurs at a faster rate than the reaction with chloroethane. [2]

- 8 (c) Many reusable face mask are made from nylon. Nylon is an example of a thermoplastic polymer.
The monomer from which the polymer Nylon 6 is produced is showed below.



- (i) Explain what is meant by the term *thermoplastic*. [1]

- (ii) Draw the structure of the Nylon 6, showing two repeat units. State the type of polymerisation involved. [2]

Type of polymerisation:

A typical washing instructions for such reusable face mask would read
“Machine wash cold, gentle cycle, do not bleach”

- 8 (c) (iii)** Suggest why the face mask should be washed in cold water during washing. **[1]**

- (iv)** With reference to the structure of nylon 6, explain why a gentle cycle should be used when washing the face mask. **[1]**

[Total: 20 marks]

End of Paper